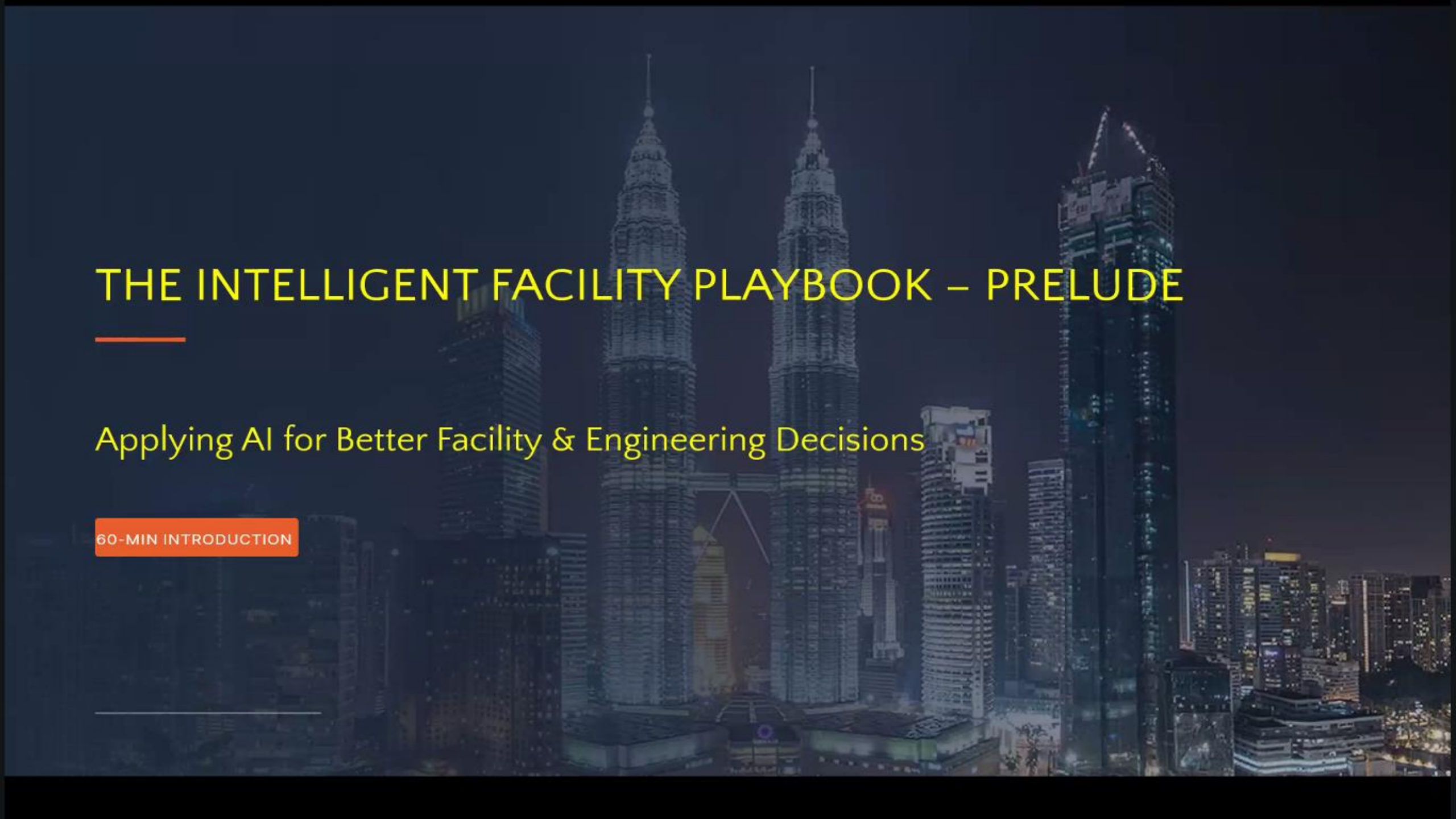




THE INTELLIGENT FACILITY PLAYBOOK – PRELUDE

Applying AI for Better Facility & Engineering Decisions

60-MIN INTRODUCTION

Current Focus

Community-centred FM, and where AI is changing how we deliver it

2021- Present Facilities Management for Social Service Agencies

A shift from purely commercial FM toward community-centred, people-first facility operations — building spaces that bring people together

USE CASE OF AI IN FM

Bridging Human Capital + Technology + AI + Space

Creating a cohesive, connected workspace — from IoT sensors and face check-in to fall detection and centre-wide dashboards for active-aging and care environments.

- Ambient conditions, people-counting & crowd density
- Contactless fall detection & panic-button alerts
- Face check-in & member attendance via face scans
- Aggregated venue heatmaps for portfolio-wide visibility



MontfortCare



We CARE
Compassion Assurance Respect Excellence



Improve Collaboration & Participation, Simplify Technology Adoption and Data Insights for Shaping Aging Policy

Various IoT

Others



People Counting



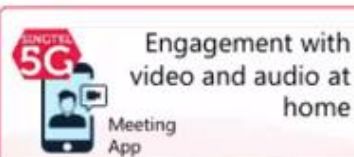
Face check-in in centres



Contactless fall detection



Panic button to alert & seek help



Engagement with video and audio at home



Engagement with video and audio at off sites



Video sharing between Aging Centres, Home & Away locations

1. Community Games
2. E-travel around the world
3. Lifestyle talk, Teleconsulting
4. Arts, Karaoke, activities

Large Format Video Canvas

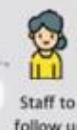


Insights from sensors & video into a single visibility across care centres, power by smart alerts

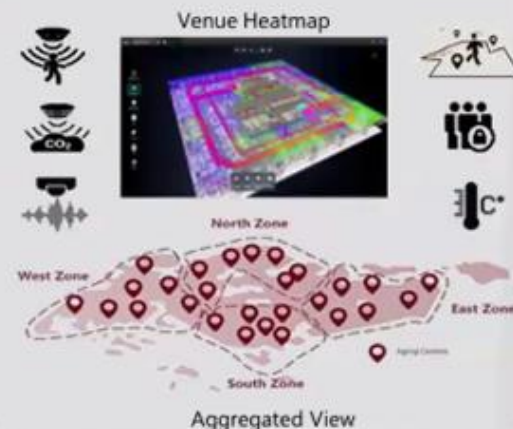
1 Observability

Trends & events based rule engine

2 Event Based Trigger



1. Centre ambient conditions such as air quality, sound level and comfort.
2. Geo-fencing, heatmap, etc.
3. People counting, crowd density and congregation.
4. Safety like fall detection, panic button.
5. Members attendance via face scans.



THE OPPORTUNITIES

Enhancing Capabilities AI in FM

DATA CENTRES IN JOHOR

The SG+ STRATEGY
Following the Singapore moratorium, the neighbouring market of Johor has seen an uptick in data centre activity in recent years.

SIZE DISTRIBUTION

Johor sees multiple hyperscale campuses that when fully completed will each deliver



>100MW

OPERATOR TYPE

96% of the market's live capacity stems from colocation facilities that are



BUILD-TO-SUIT

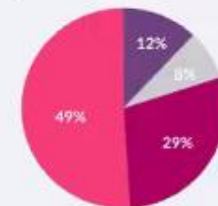


CLUSTERS

Data centres have emerged in more than one cluster in this emerging market. Beyond the city centre, data centres are noted to cluster around the Nusajaya Tech Park, Sedarak Tech Park, YTL Green Data Centre Park and Nusa Cemerlang Industrial Park.

TOTAL CAPACITY

The majority of the market's capacity of nearly 1,600MW is in the development pipeline, including projects under construction and those which have or are in the midst of securing the necessary elements (land, power, government approvals) for development.



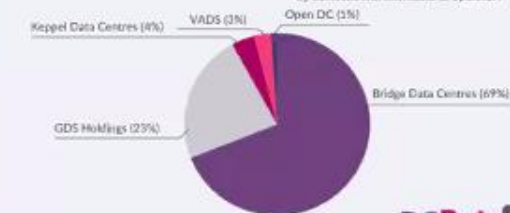
TOTAL SUPPLY

In the past five years, the total supply (live, under construction, committed and early stage) of data centres in Johor have grown by 100 times.



MARKET SHARE BY LIVE CAPACITY

Live capacity in the Johor market is presently fueled by domestic and international operators.



DCByte

Fueling your data centre strategy with trusted intelligence | dcbyte.com

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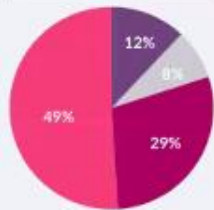


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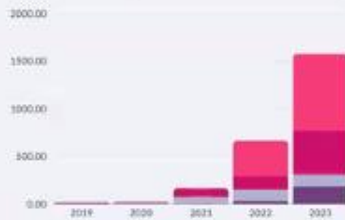
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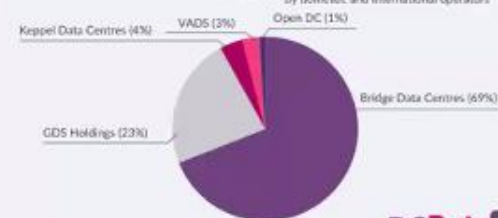
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Malaysia's Growing Facilities Opportunities

THE EDGE MALAYSIA

Home Corporate Economy Court & Politics Sections Edge

Economic Focus

Malaysia poised to capture major real estate growth amid region's structural transformation — JLL

By Bernama 04 Aug 2026, 01:45 pm

"Johor is also emerging as a potential global data centre hub, capitalising on Malaysia's growing role as a regional digital infrastructure centre," it said in a report titled, "Planning for a Multipolar World: Southeast Asia — Poised for Growth".

JLL added that Penang's Silicon Island reinforces the state's critical role in high-value industries, while adjacent states, including Kedah and Perak, are well-positioned to benefit from spillover demand arising from land constraints.

Meanwhile, it said improved rail connectivity in Greater Kuala Lumpur is broadening the catchment area for industrial and township development, while the capital advances into innovative sectors, including aerospace.

JLL head of research and advisory for Southeast Asia Dr Yang Liang Chua said Malaysia exemplifies the shift taking place across Southeast Asia, from single-city concentration to networked, corridor-led growth.

"The Johor-Singapore axis represents not just cross-border cooperation, but a fundamental reimagining of how economic activity can be distributed across interconnected systems," he said.

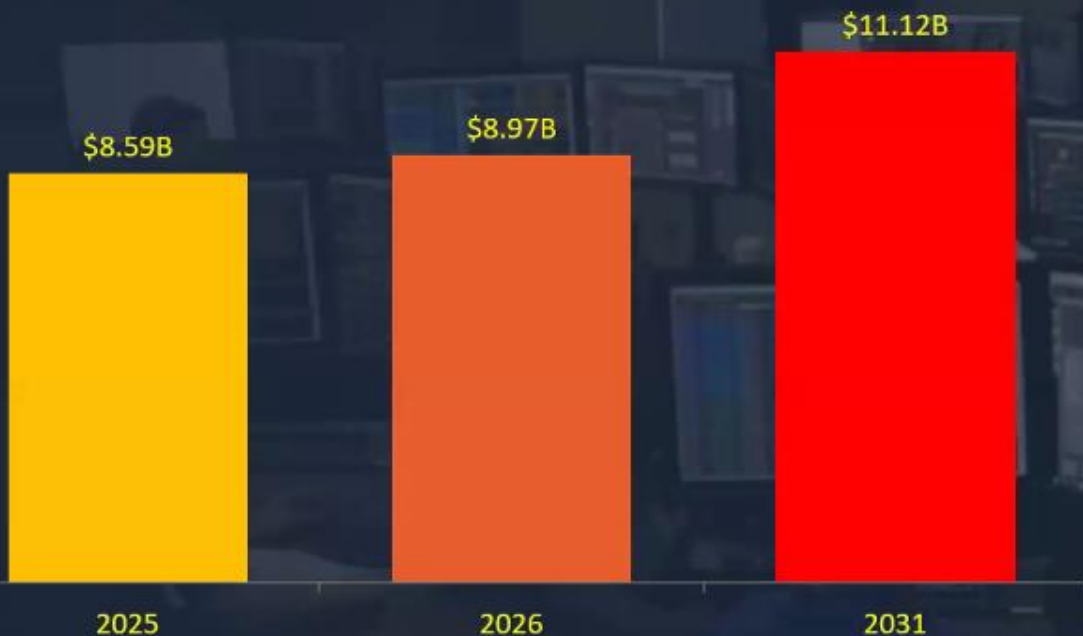
JLL Malaysia head of research and advisory Yulia Nikulicheva said Malaysia's diversified ecosystem makes the country attractive to major international players seeking long-term investments and risk diversification in the region.

Malaysia's growth trajectory is increasingly being defined by strategic cross-border and industrial linkages, with several developments positioning the country at the forefront of regional real estate opportunities.

Malaysia's AI Opportunity

A fast-growing market, and a fast-accelerating adoption curve

MALAYSIA FACILITY MANAGEMENT MARKET (USD BILLION)



CAGR 4.39% • Fuelled by data-centre growth in Johor • Source: Mordor Intelligence

MALAYSIA FM IS MOVING TOWARD

**Digital twins • Energy monitoring dashboards •
Predictive infrastructure alerts • AI-driven CMMS &
workflow automation**

ARE WE READY TO RIDE THE AI WAVE?

24

Single digits

**AI adoption across corporate properties —
limited by legacy spreadsheets and manual
logs**

25

+35% YoY

**The Infrastructure Boom — corporate AI
adoption across Malaysian businesses surged**

26

1.9% → 4.0%

**Job postings requiring specialised AI skills
doubled in 24 months as firms scramble to
hire tech-literate engineers**

Closing the Gap

Where the Malaysian mid-tier FM market is stuck, and how AI moves it forward

THE GAPS

Fragmented Data

Older buildings lack digital blueprints or structured asset registries — info trapped in binders and siloed spreadsheets.

Engineering Skill Shortage

Few engineers combine traditional building mechanics (HVAC, chillers, switchgear) with data science/IoT skills.

Upfront Capital Resistance

Owners see FM as pure expense; smaller vendors lack cash flow for expensive software deployments.

Labor Over-Reliance

Cheap manual labor has disincentivized mid-tier firms from investing in workflow automation.

WHERE AI HELPS

AI acts as the bridge that converts raw, chaotic property data into automated operational actions.

- Predictive Chiller & HVAC Diagnostics
- Automated Generative Work Orders
- Dynamic Energy Load Profiling
- Automated Workflows & Live Dashboards
- CAPEX & OPEX Planning Support

AI is a triage & efficiency layer — engineering judgement and sign-off remain with the qualified professional.



01

SETTING THE CONTEXT

Mindset, objectives, and how this session works



Regulatory Compliance

Contractor Performance

Equipment alarms

Tenant complaints

Rising energy bills

**Have you ever made a decision that
looked right.....but turned out to be
completely wrong?**

**The Reality of Facilities Management - Every day, Facility
Managers make decisions with incomplete information.**

Objectives



Structured Thinking

Use AI with structured thinking frameworks for better analysis



Better Decisions & Communicate Better

Make better decisions with AI support



By the end of this workshop, you will have practical frameworks to understand AI outputs.

The Right Mindset for This Workshop

 This is NOT a prompting session

WHAT WE FOCUS ON



Thinking

Structured approaches to analyse engineering data



Validation

Verify AI outputs against engineering reality



Decisions

AI supports engineering judgment — it doesn't replace it



Key Principle: AI is a tool. Your expertise validates the output — never the other way around.

02

AI IN FACILITIES OPERATIONS

What AI can and cannot do in your daily work

LEGEND	
SYMBOL	DESCRIPTION
	VOLUME CONTROL DAMPER
	FIRE DAMPER
	MOTORISED DAMPER
	RAIL RECOVERY WHEEL
	PRE-COOLED FAN UNIT
	PRE-COOLED AIR HANDLING UNIT
	SUITED FAN COIL UNIT
	CABINET FAN COIL UNIT
	MALL AIRSPREAD FAN COIL UNIT
	OUTDOOR COOLING UNIT (WITH REFRIGERANT TYPE)

Context -

- Set of BMS Data of chiller & Cool tower
- Grp Comprise of DOE & CE from the region
- Objective using any AI tool to analysis the data given

Data was spike with false reading

What AI can and cannot do in your daily work

Example of data set – Chiller data & Cooling tower

Timestamp	Chiller_Load_%	Power_kW	COP_Reported	CHW_Supply_C	CHW_Return_C	CW_Inlet_C	CW_Outlet_C	Flow_m3h
2025-01-01 0:00:00	35	485.6	4	6.8	11.7	31.6	38.5	632
2025-01-01 0:15:00	69	699.6	8.66	6.7	10.8	32.1	37	680
2025-01-01 0:30:00	34	275.7	6.12	7	12.2	30.7	36.3	629
2025-01-01 0:45:00	79	944.3	5.01	7.1	12.8	32.5	37.2	594
2025-01-01 1:00:00	30	300.3	5.07	6.3	11.3	32.7	36.9	646
2025-01-01 1:15:00	71	633.5	5.52	6.7	13.4	29.3	35.8	561
2025-01-01 1:30:00	56	658.2	7.24	6.5	7.8	32	37.4	683
2025-01-01 1:45:00	71	775.9	6.29	7.5	13.8	29.4	34	471
2025-01-01 2:00:00	45	368.3	5.6	6.5	12.9	29.3	34.1	572
2025-01-01 2:15:00	75	693.5	5.14	6.7	12.1	31	35.6	410
2025-01-01 2:30:00	77	1081	4.45	6.6	13.5	30	34	406
2025-01-01 2:45:00	35	373.5	6.4	6.6	13	30.6	35.6	460
2025-01-01 3:00:00	75	894	5.39	6.6	11.7	32.8	37.4	668
2025-01-01 3:15:00	64	553.2	6	7.5	13.3	30.6	37.2	569
2025-01-01 3:30:00	56	607.1	5.9	6.5	13.1	30.8	37.4	437
2025-01-01 3:45:00	64	558.2	5.05	7.9	11.9	32.9	39.5	414
2025-01-01 4:00:00	48	406.1	5.53	6.1	10.5	31.1	32.8	472
2025-01-01 4:15:00	42	479.6	5.64	7.5	13.5	32.1	39	611
2025-01-01 4:30:00	30	333.3	6.17	7.3	12.8	30.5	35.3	669
2025-01-01 4:45:00	63	926.9	4.45	6.8	11	32.2	38.5	562
2025-01-01 5:00:00	60	583.3	5.36	6.2	12.4	29.1	30.6	633
2025-01-01 5:15:00	47	467.5	5.6	7.1	13.4	32.1	33.6	499
2025-01-01 5:30:00	87	731.9	9.13	7.7	14.2	31.3	37.1	514
2025-01-01 5:45:00	72	821.6	6.08	6.9	11.5	29.6	30.8	423
2025-01-01 6:00:00	34	360.2	5.24	6.9	12.8	29.8	36.3	444
2025-01-01 6:15:00	40	425.4	5.97	8	13.8	32.6	37.2	643
2025-01-01 6:30:00	59	571.4	5.1	7.8	13	32	39	509
2025-01-01 6:45:00	35	388.4	6.08	6.4	11.6	32.5	37.6	619
2025-01-01 7:00:00	68	759.5	5.07	7	12.2	30.3	35.1	592
2025-01-01 7:15:00	40	456.5	3.65	7	13.6	32.1	39	580
2025-01-01 7:30:00	66	607.6	5.35	7.4	13.8	31.4	37.9	502
2025-01-01 7:45:00	66	714.1	8.55	6.8	12.9	32	36.9	652
2025-01-01 8:00:00	57	508.9	10.21	6.8	12.2	30	36.1	604
2025-01-01 8:15:00	73	618.7	6.23	6.2	13	33	34.8	582
2025-01-01 8:30:00	53	497.9	5.57	6.5	13.1	30	35	620
2025-01-01 8:45:00	58	499.2	6.85	7.8	9.6	29.8	35.6	471
2025-01-01 9:00:00	51	607.8	6.15	7.9	13	32.7	37.3	582
2025-01-01 9:15:00	67	628.6	5.04	8	14.7	29.6	36.6	612

Timestamp	Ambient_C	WetBulb_C	CT_Inlet_C	CT_Outlet_C	Approach_C	Fan_kW	Water_Use_m3	Blowdown_m3
2025-01-01 0:00:00	30.3	23.9	35.4	30.8	6.9	53.3	16.9	2.1
2025-01-01 0:15:00	31.5	24.5	37.4	34.9	10.4	38	5.5	1.3
2025-01-01 0:30:00	34.7	30.6	34.4	30.2	-0.4	23.2	19.4	3.4
2025-01-01 0:45:00	32.3	27.1	34.7	28.2	1.1	20.2	13.7	4.8
2025-01-01 1:00:00	32	27.9	34.6	30.5	2.6	37.1	19.8	4.9
2025-01-01 1:15:00	31.9	27.7	35.7	31.5	3.8	25.3	14.7	4.9
2025-01-01 1:30:00	32.6	26.2	33.2	28	1.8	38	15.5	3.9
2025-01-01 1:45:00	31.2	24.3	34	28.3	4	20.4	21.3	2.1
2025-01-01 2:00:00	33.9	28.8	35.1	29.8	0.9	37	15.9	2.7
2025-01-01 2:15:00	31.5	26.5	36.4	29.9	3.4	55.9	22.2	3.2
2025-01-01 2:30:00	31.3	25.3	36.1	30.3	5	25.8	18.3	4.6
2025-01-01 2:45:00	33.6	28.4	35.7	30.9	2.5	26.1	13.3	2.1
2025-01-01 3:00:00	31.5	25.7	36.9	32.8	7.1	31.6	24.9	1.1
2025-01-01 3:15:00	32.3	26.6	37.3	31.6	5	31.1	22.8	4.7
2025-01-01 3:30:00	30.3	24.1	36.3	31.7	7.6	37.2	18.2	3.7
2025-01-01 3:45:00	34.3	29.3	34.4	29.7	0.4	60.9	20.4	2.8
2025-01-01 4:00:00	33.3	26.6	34.3	32.5	5.9	26.5	7.7	2.4
2025-01-01 4:15:00	34.5	28.1	35.8	29.1	1	54.2	24.4	3
2025-01-01 4:30:00	33.7	28.4	34.3	28	-0.4	21.3	13.7	1.2
2025-01-01 4:45:00	30.2	25.3	37	31.5	6.2	30.1	19.7	3.5
2025-01-01 5:00:00	33.4	28	35.2	30.9	2.9	22.7	13.6	4.1
2025-01-01 5:15:00	33.4	28.7	33.2	28.1	-0.6	31.9	20.4	4
2025-01-01 5:30:00	31.9	26	35.8	30.6	4.6	25.3	12.5	1.5
2025-01-01 5:45:00	31.5	27	36.3	31.5	4.5	22.6	12.7	1.7
2025-01-01 6:00:00	32.1	27.6	33.6	28.9	1.3	21.4	12.5	3
2025-01-01 6:15:00	33	28.2	35.5	29.3	1.1	29.9	16.6	2.4
2025-01-01 6:30:00	32.2	28	34.6	29.7	1.7	30.3	18.7	2.5
2025-01-01 6:45:00	31.6	26.2	34.3	28.3	2.1	26	12	1.8
2025-01-01 7:00:00	30.3	24	33.3	26.5	-0.6	31.3	23.2	3.5
2025-01-01 7:15:00	33.1	28	35.3	29.7	-0.3	20.1	15.8	2.1
2025-01-01 7:30:00	32.7	27	37.4	32.9	5.9	34.3	12.9	3.2
2025-01-01 7:45:00	31.8	25.4	34.9	28.9	3.5	57.1	13.4	1.9
2025-01-01 8:00:00	31.8	27.3	36.4	32	0.6	23.4	22.6	3.2

What Happens When Nobody Checks – Case Study

COP – (Coefficient of Performance)
reading of 9.5 —
looked like excellent
performance

A cooling tower
approach
temperature of 0°C —
physically impossible

High power draw
paired with a high ΔT
— quietly pointed to
fouling

What is the Cost to Your Management ?

Overstated performance → no retrofit
approved → inefficiency continues →
higher operating cost, every month,
indefinitely

Wrong sensor data → false
optimisation → system instability →
occupant complaints → emergency
repairs.

Engineer A gets the AI-generated performance report. It looks clean. He signs off.



Engineer B gets the same report. She asks four questions before she signs anything.



SIX MONTHS LATER...

Engineer A is explaining to management why the "optimized" chiller plant needs emergency repairs.



Engineer B is the one who caught it before it shipped.



Same tools. Same data. Different outcome — because of what happened between "AI said so" and "I signed off."

The Choice:
Before vs After

What AI Can & Can not Do ?

AI Can

- ✓ Analyse trends
- ✓ Summarise reports
- ✓ Detect anomalies
- ✓ Draft recommendations

AI Can't

- ✗ Understand design intent
- ✗ Understand Operational Context
- ✗ Smell overheating insulation
- ✗ Hear abnormal vibration
- ✗ Verify sensor calibration
- ✗ Accept legal accountability

AI analyses. FM validate.

03

STRUCTURED THINKING FRAMEWORKS

Human-led AI validation and decision-making models

The Core Problem



AI gives answers. But are they correct?

Are they safe to implement in your building?

THE TRUST GAP



Did the AI understand your building's unique constraints?



Did it verify sensor accuracy before making recommendations?



Did it account for local regulations and utility rates?

The answer to all three is NO. That's why you need structured validation frameworks.

Understanding AI Risks



Hallucinations

AI generates plausible-sounding but factually incorrect information



Bias

Training data biases lead to skewed recommendations



Data Issues

Garbage in, garbage out – flawed inputs produce flawed outputs



Overconfidence

AI presents uncertain conclusions with false certainty



Confident AI + Bad Data = Wrong Decisions

Wrong emission factors alone can overstate or understate carbon impact by 30–50%.
Your validation is the only thing standing between this equation and your building.

Building Human-AI Partnership

What Humans Bring

- Domain expertise
- Operational context
- Risk judgment
- Stakeholder management
- Accountability



HUMAN-AI PARTNERSHIP

Engineering expertise + AI analytical power = Better decisions

What AI Brings

- Pattern recognition
- Speed of analysis
- Data processing scale
- Consistency
- 24/7 availability

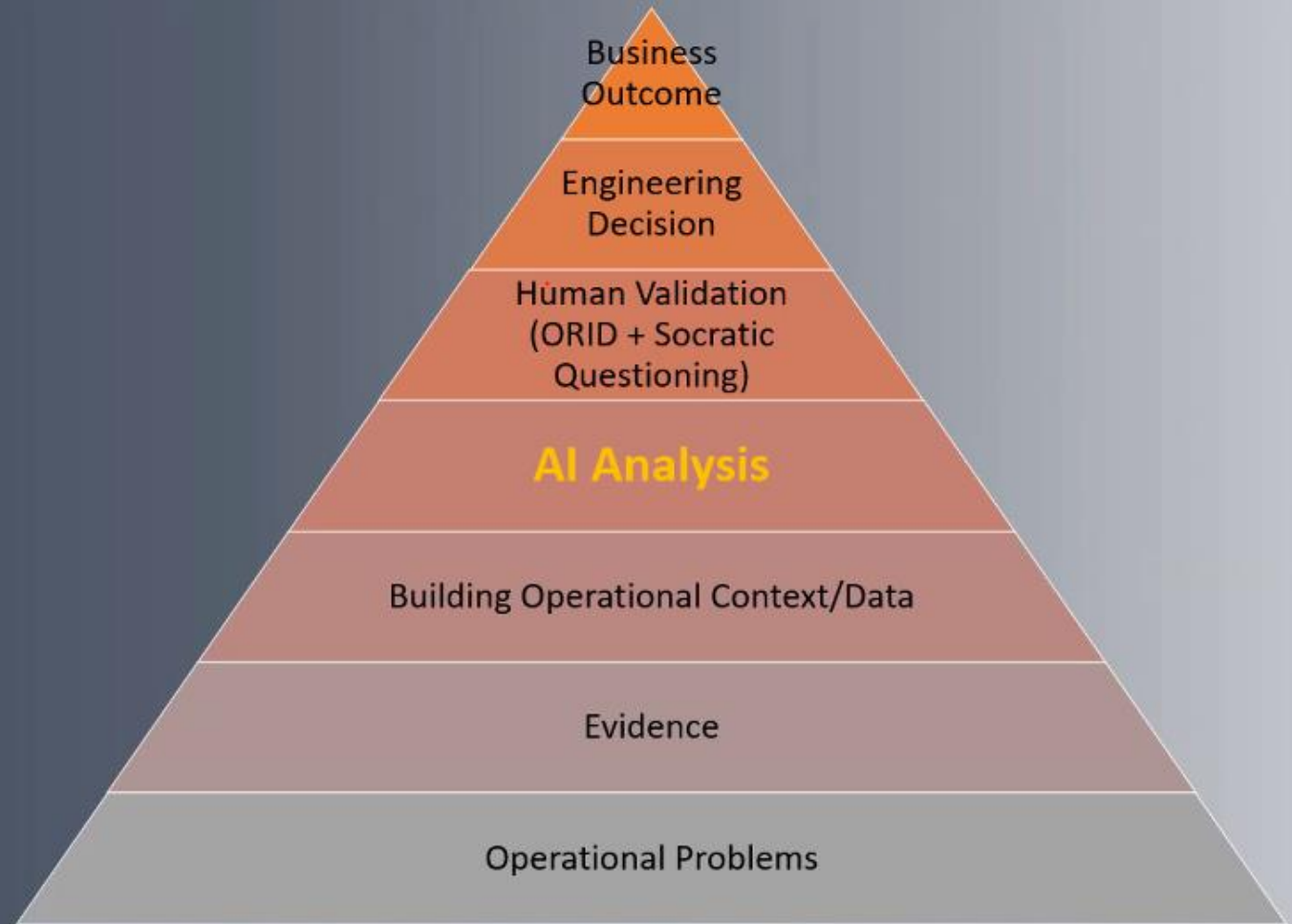
An empowered human-AI partnership creates **readiness** for deeper capability development programs.

The Intelligent FM Decision Framework™

AI WILL NOT REPLACE THE FACILITY MANAGER.

AI WILL REPLACE FACILITY MANAGERS WHO CANNOT THINK CRITICALLY.

THE FACILITY MANAGERS WHO CAN COMBINE OPERATIONAL DATA, JUDGEMENT AND AI WILL BECOME MORE VALUABLE.



Combining ORID + Socratic Questioning

ORID guides your thinking. Socratic questioning protects your decisions.

OBJECTIVE

REFLECTIVE

INTERPRETIVE

DECISIONAL

SOCRATIC QUESTIONS AT EVERY STAGE

At Objective: What evidence supports these numbers? Are the sensors reliable?

At Interpretive: What assumptions is the AI making? What could be wrong?

At Reflective: Is this realistic? Does it match my experience?

At Decisional: What would I verify before implementation?

The **ORID framework** and **Socratic Questioning** work in tandem as a human-led validation model where ORID provides the structure for thinking, and Socratic Questioning provides the rigour to protect decisions. In facilities management, this combination ensures that the engineer's expertise always validates the data output, preventing costly mistakes caused by blind trust in AI or faulty sensors

The Intelligent Facility Manager

Bridging Engineering, AI and Business Decisions

TECHNICAL SIDE



- Failures
- Sensors
- Maintenance
- Equipment

INTELLIGENT FACILITY MANAGER + AI

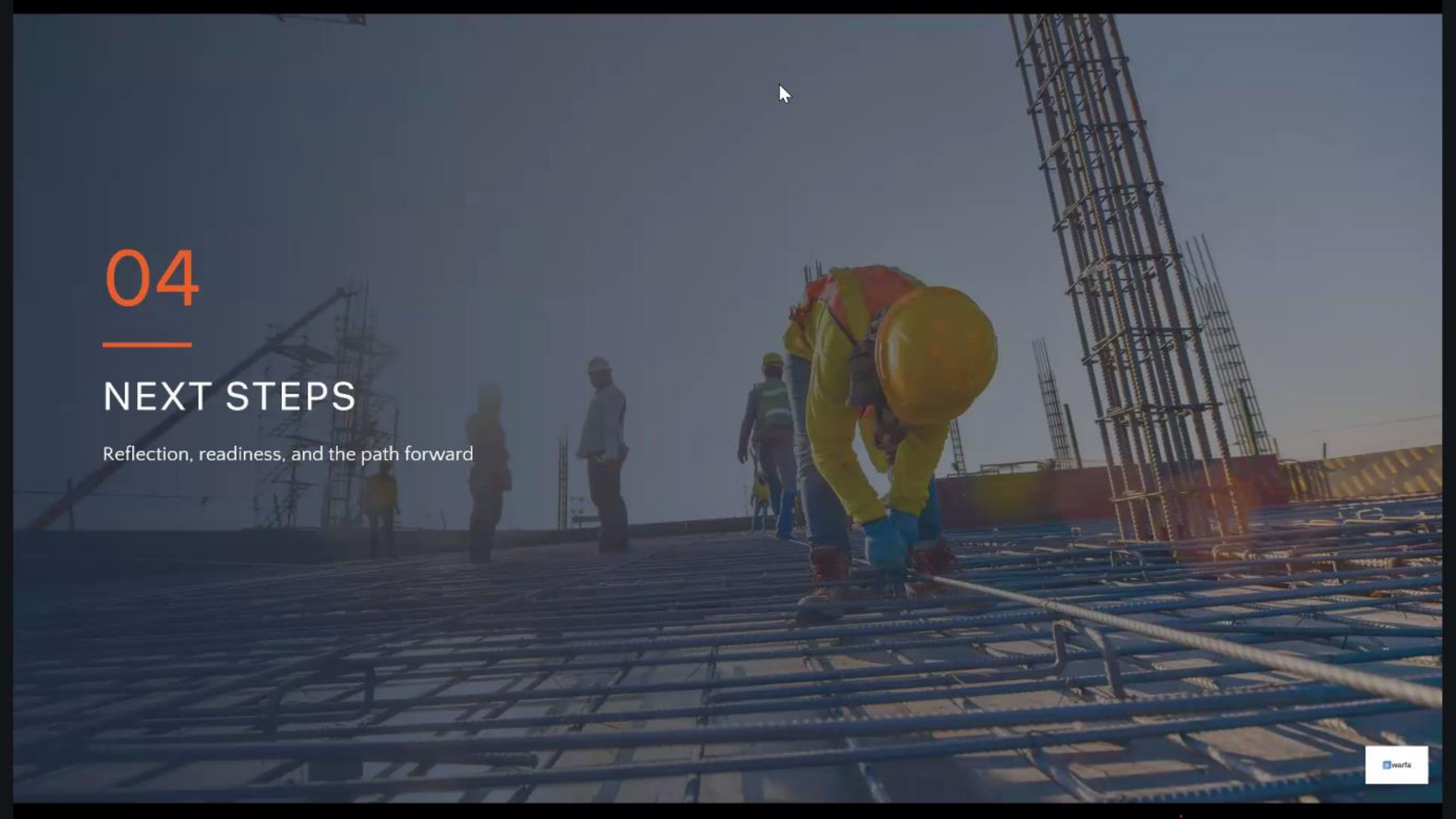


- Technical expertise earns credibility.
- Engineering judgement builds trust.
- Clear **COMMUNICATION** enables decisions.
- **AI** simply makes all three more powerful

MANAGEMENT SIDE



- Business Goals
- Budget
- Risk
- Stakeholders

A construction worker in a yellow safety suit and hard hat is working on a steel rebar grid at a construction site. The worker is in the foreground, bent over, adjusting the rebar. In the background, other workers are visible, and tall rebar structures rise against a clear sky. The scene is set during the day with bright lighting.

04

NEXT STEPS

Reflection, readiness, and the path forward

HRD CORP-ALIGNED • 2-DAY CLASSROOM PROGRAMME

The Intelligent Facility Handbook

AI-Enabled Facilities Management for the FM Professional

- **Analyse operational data.**
- **Validate AI recommendations.**
- **Lead & communicate , compliant decisions.**

By the end of this programme, you'll analyse real facilities data, validate AI-generated recommendations with structured engineering judgement, and walk away with an AI-assisted FM workflow built for your own organisation.

2 DAYS

14 HOURS
CLASSROOM

CAPSTONE PRESENTATION

HRD CORP CLAIMABLE

Your 4-Step Learning Journey

Bridging the gap between traditional expertise and emerging AI workflows

1

Build Your FM Foundation

Ground yourself in modern FM principles, and see exactly how Generative AI acts as a force multiplier — never a replacement for your technical judgement.

2

Analyse Real Operational Data

Read CMMS records, BMS trends, energy consumption, work orders and asset data like an expert catching performance issues and risks before they escalate.

3

Validate AI-Assisted Recommendations

Apply ORID and Socratic Questioning to stress-test AI-generated recommendations before they touch an operational, maintenance or compliance decision.

4

Make Better Engineering Decisions

Turn evidence into recommendations you can defend — communicated with confidence to management, clients and contractors through AI-assisted reports.

What You'll Walk Away With

Real skills. A real deliverable. A workflow you can use

YOUR TAKEAWAYS

- **A working FM Workflow**
built for your own organisation — not a generic template
- **A capstone presentation**
to showcase your workflow and defend it in front of peers
- **A validation habit that sticks**
ORID + Socratic Questioning, applied to any AI output you meet
- **Confidence at the management table**
present evidence-based recommendations to clients & contractors

WHO SHOULD ATTEND

FM Managers, Building Executives, M&E Engineers and Facility Coordinators in Malaysia who oversee CMMS/BMS operations, statutory compliance, or day-to-day facility decisions.

No coding or data-science background required.

SEATS ARE LIMITED

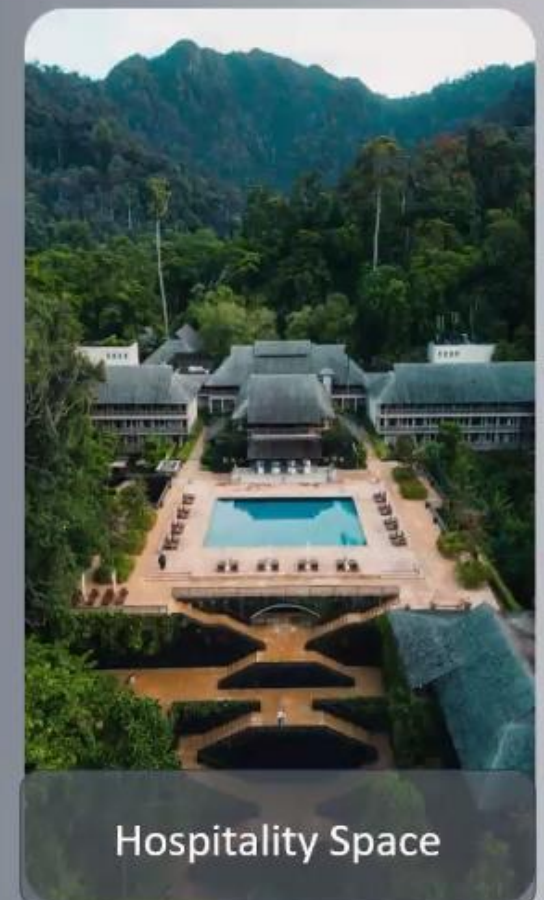
Secure Your Seats

Dates, venue & fees: TBC | HRD Corp claimable

Register: [insert email / contact]

Built For Every Type of Facility

Different buildings. Same physics. Same risk of blindly trusting an AI output — unless you know how to check it.



It is a Human-AI Partnership



“You won’t lose your job to AI you’ll lose it to your coworker who uses it”

Nvidia CEO Jensen Huang

Over the next five years, the most valuable FM professionals will be those who combine Facilities **engineering expertise, AI literacy, data-driven decision-making, and business leadership.**

AI becomes a force multiplier, not the decision-maker while accountability remains with the engineer and FM leader